H

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE NAME	**ANSWERS & COMMENTS**

BIOLOGY CLASS	2bi2___	REGISTRATION NUMBER	

BIOLOGY **9744/02**

Paper 2 Structured and Free-response Questions **23 August 2022**

Candidates answer on the Question Paper. **2 hours**
No Additional Materials are required

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/10
2	/10
3	/10
4	/10
5	/10
6	/10
7	/10
8	/10
9	/10
10	/10
Total	/100

This document consists of **12** printed pages.

1 Fig. 1.1 shows the reproductive cycle of human herpesvirus (HHV).

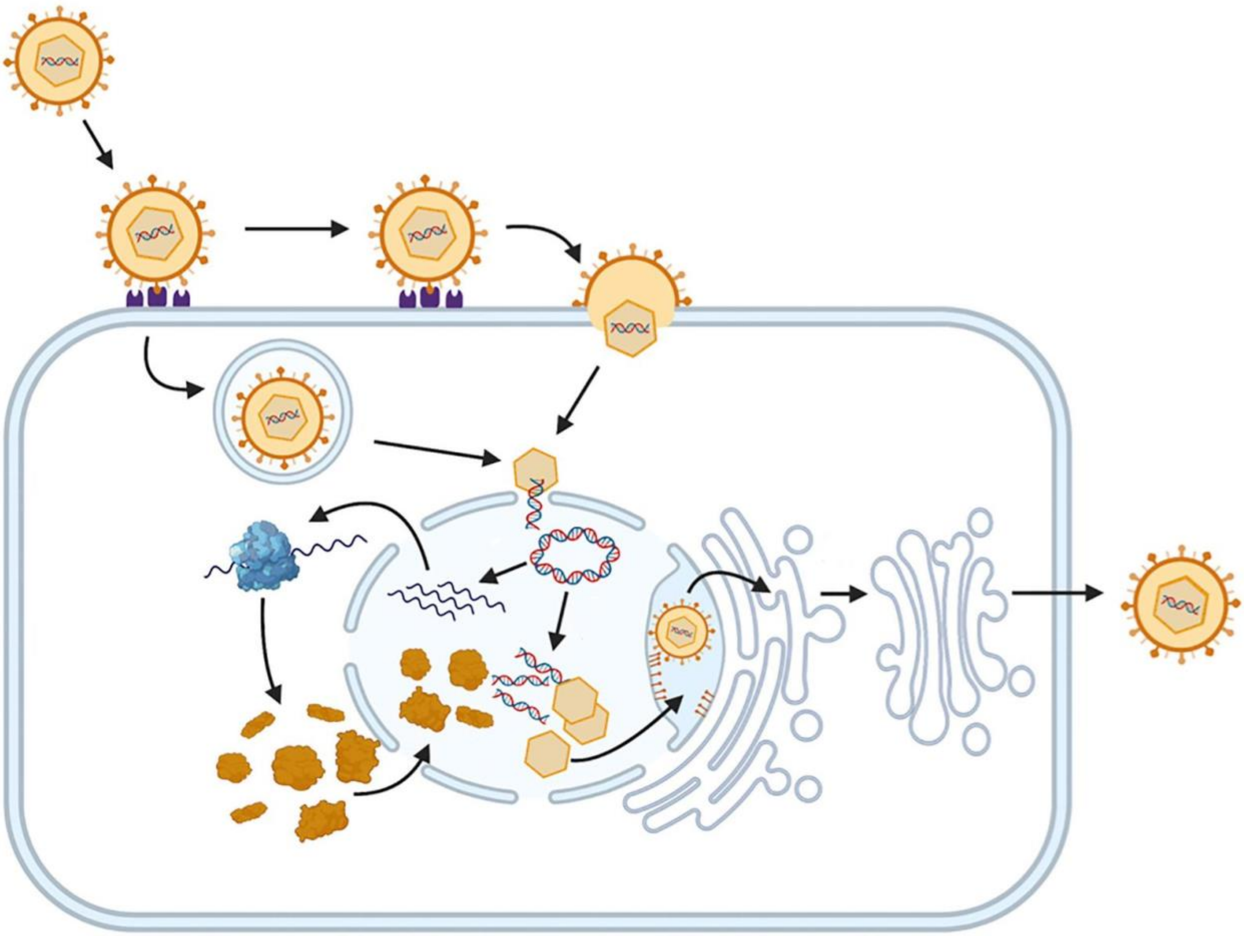


Fig. 1.1

(a) With reference to Fig. 1.1, describe the structure of HHV.

LO 1(e); AOA [3]

Answers:

1. (one copy of) double-stranded / helical / linear / non-segmented genome / nucleic acids / DNA / RNA;
2. icosahedral / hexagonal capsid / protein coat;
3. (viral / lipid / membrane) envelope / surface / phospholipid bilayer with (different) glycoproteins / spike proteins / antigens;
4. overall shape of virion: spherical / circular / round;

Comments:

(b) Name **two** membranous organelles that are shown in Fig. 1.1 and outline their roles in the reproductive cycle of HHV.

LO 1(b); AOB [4]

Answers:

1. nucleus;
2. site of viral genome replication / viral gene transcription / assembly of virions;
3. (rough) endoplasmic reticulum;
   (reject: undefined abbreviation; smooth endoplasmic reticulum)
4. transports newly assembled virions to Golgi body / apparatus;
   (reject: site of viral protein synthesis / modification)
5. Golgi body / apparatus;
   (reject: golgi)
6. transports newly assembled virions to host cell surface membrane;
   (reject: site of viral protein modification)
7. endosome / endocytic vesicle;
8. forms to enable entry of HHV into host cell;

(reject: ribosome)

maximum 2 marks for naming membranous organelles

maximum 2 marks for outlining respective role of named membranous organelles

Comments:

(c) Discuss how viruses such as HHV challenge the cell theory and concepts of what is considered living.

LO 1(f); AOB [3]

Answers:

1. ref to cell theory that “living organisms are composed of cells”;
2. ref to one argument to support “viruses are considered non-living”;
3. ref to one argument to support “viruses are considered living”;

Comments:

[Total: 10]

2 **(a)** Fig. 2.1 shows the structure of the amino acid cysteine.

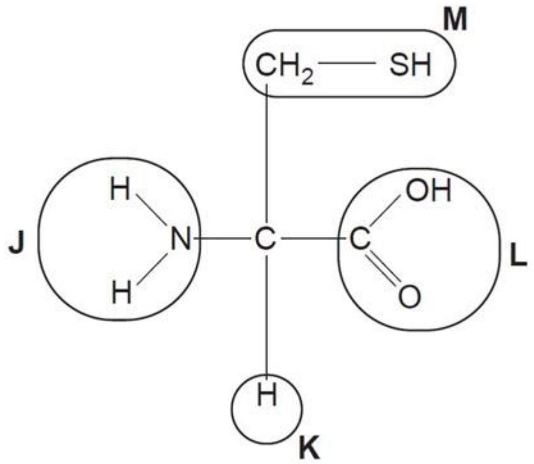


Fig. 2.1

(i) Complete the table by selecting the letter, **J**, **K**, **L** or **M**, that represents each of the following groups in cysteine.

group	letter
carboxyl	**L**
amino	**J**

LO 1(g)(iii); AOA [1]

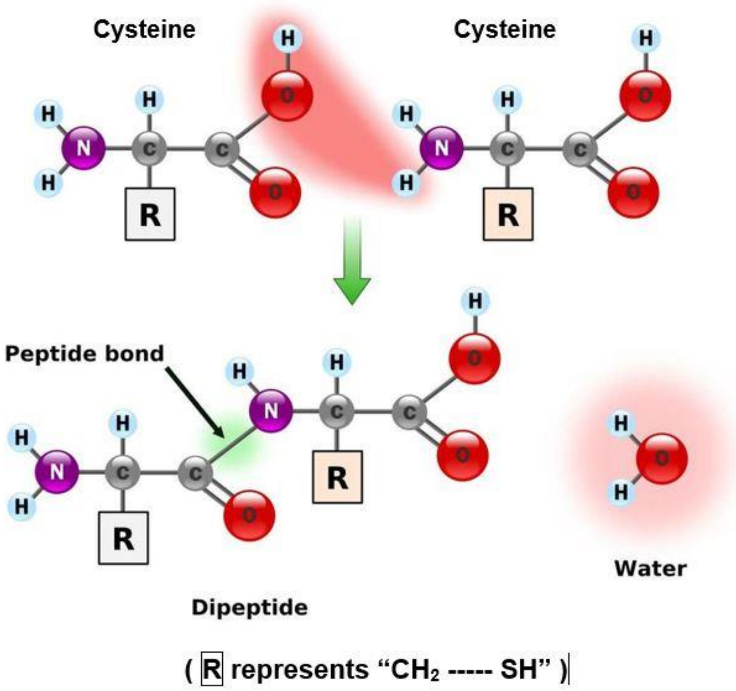
Comments:

(ii) The primary structure of a protein consists of a chain of amino acids.

With reference to Fig. 2.1, draw how two cysteines would be joined together in forming the primary structure of a protein. Label and name the bond(s) formed.

LO 1(h)(iii); AOA [3]

Answers:



1. Both cysteines drawn have the same structure as shown in Fig. 2.1;
2. Peptide bond / linkage formed is correctly labelled and named;
3. H (from amino group) combines with OH (from carboxyl group);
4. condensation reaction stated OR water molecule drawn as one of the products;

Comments:

(iii) Describe the role of group **M** in the protein structure.

LO 1(m); AOA [1]

Answers:

1. allows <u>disulfide bond(s) / linkage(s) / bridge(s)</u> to be formed between sulfhydryl (-SH) functional groups in **M**, thereby stabilising <u>tertiary structure and/or quaternary structure</u> of the protein;

Comments:

(b) Pepsin is a protease found in the stomach.

Fig. 2.2 shows the effect of varying substrate concentration on pepsin activity.

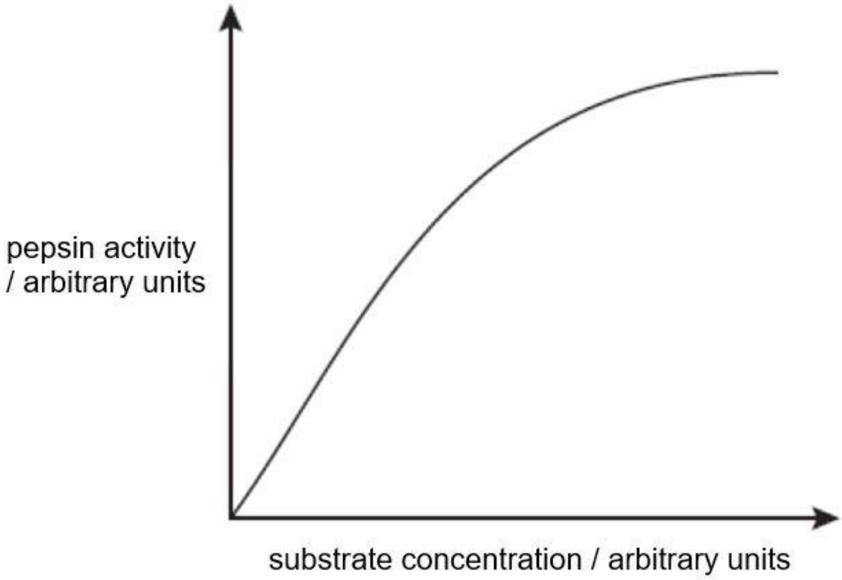


Fig. 2.2

(i) Pepstatin is known to be a non-competitive inhibitor of pepsin.

Explain the effects of pepstatin on pepsin.

LO 1(r), 1(s); AOB [3]

Answers:

1. Pepstatin has no close structural resemblance to the (protein) substrate of pepsin;
2. Pepstatin binds to a region other than active site of pepsin and induces change in its globular three-dimensional conformation altering its active site;
3. (Protein) substrate is still able to bind to the active site but catalysis cannot take place due to changes in the nature of the catalytic groups at the active site;
4. This puts a proportion of pepsin out of action, lowering rate of pepsin activity. Increasing the substrate concentration does not increase rate of reaction;

Comments:

(ii) On Fig. 2.2, draw a graph to show how the activity of pepsin would vary with its substrate concentration in the presence of pepstatin. Annotate the maximal velocity as 'V_{max} (inhibited)' on the graph that you have drawn.

LO 1(s); AOB [2]

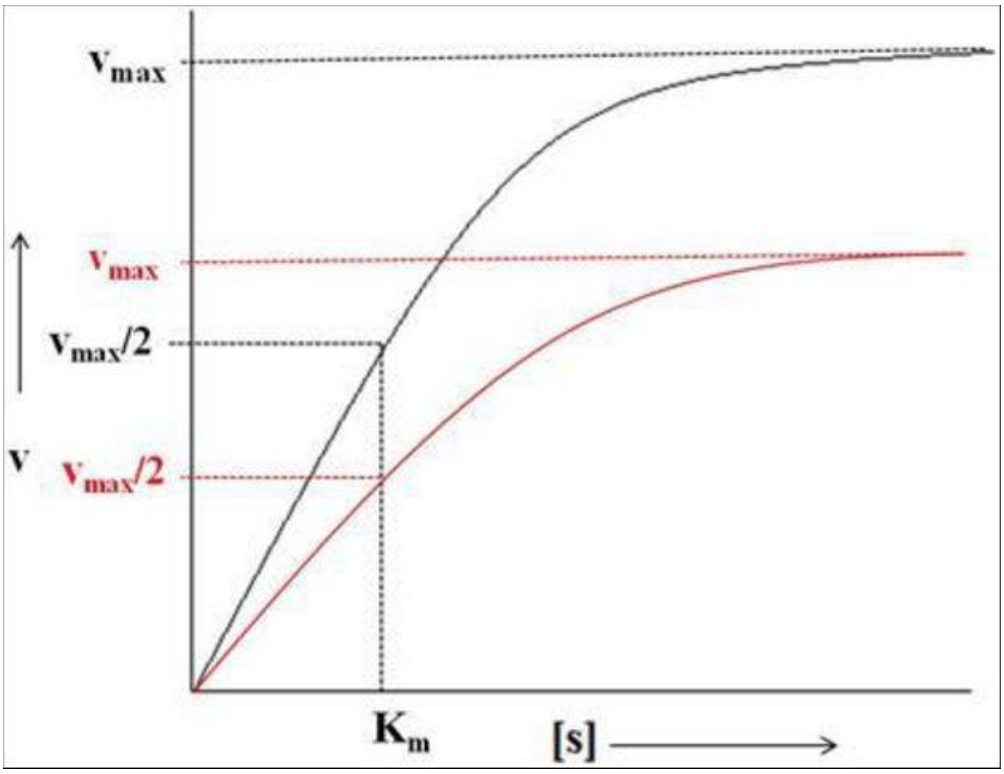
Answers:

1. correct graph drawn, i.e. V_{max} (inhibited) < V_{max} (uninhibited);

2. V_{max} (inhibited) annotated correctly on the vertical axis;

(No penalty for not drawing same K_m for both uninhibited and inhibited reactions.)

(Maximum 1 mark for this question if never draw on Fig. 2.2)



Comments:

[Total: 10]

3 Fig. 3.1 shows the percentage of rat cells undergoing DNA replication. Some cells contained a protein called cyclin D and some cells did not. All cells were in early interphase at time 0.

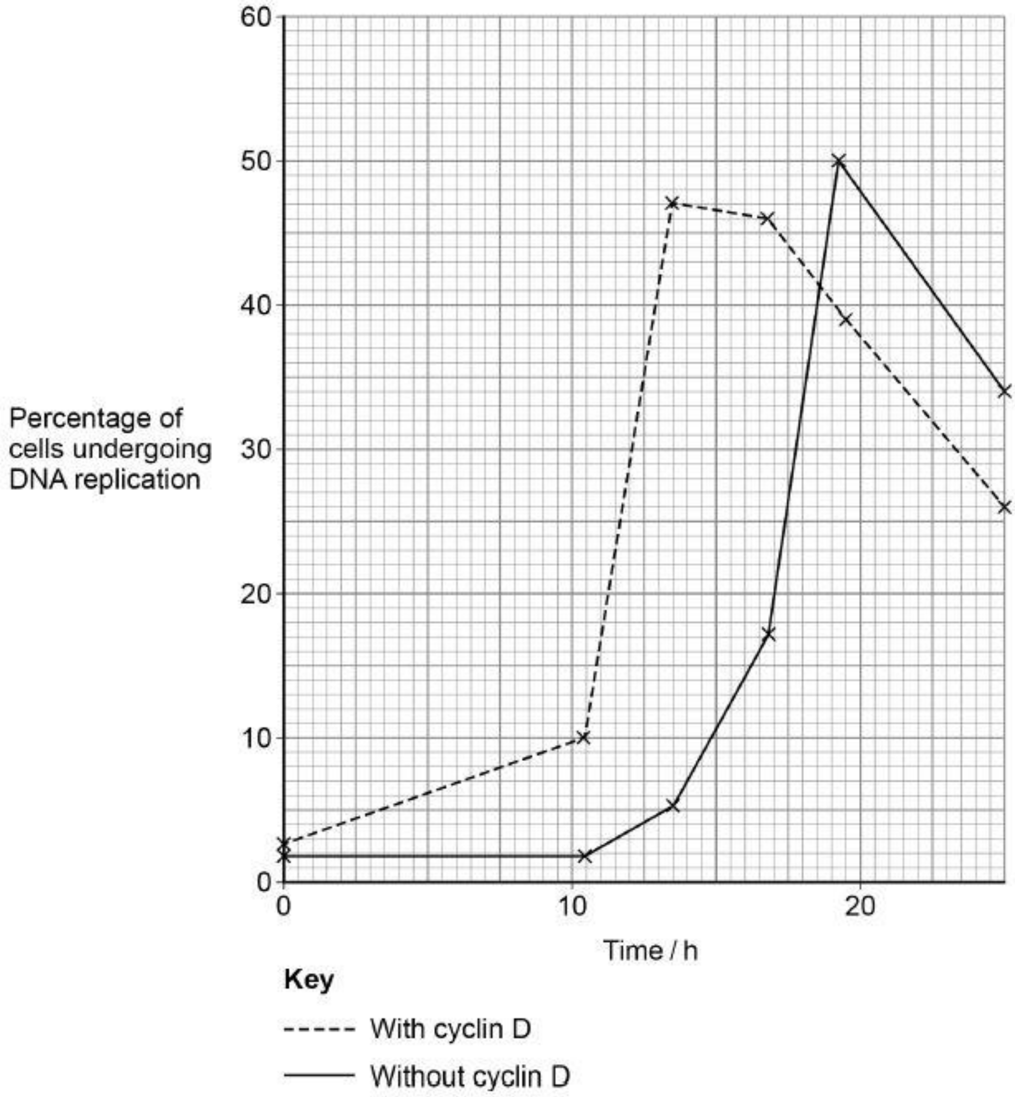


Fig. 3.1

(a) Name the stage in early interphase when DNA replication occurs.

LO 2(n); AOA [1]

Answers:

1. Synthesis / S phase;

Comments:

(b) With reference to Fig. 3.1, describe the effect of cyclin D on cells.

AOB [2]

Answers:

1. quote values from both axes to show that there is a higher percentage of cells undergoing DNA replication at an earlier time point for cells with cyclin D;
2. decreases the time needed for cells to be undergoing DNA replication OR increases the rate at which cells enter S phase;

Comments:

(c) Cyclin D is known to bind specifically to a cyclin-dependent kinase (Cdk), which then acts on DNA polymerase.

Suggest how cyclin D may cause a change in the activity of DNA polymerase.

LO 1(p); AOB [2]

Answers:

1. Cyclin D binds to Cdk, which then undergoes a conformational change that activates the Cdk;
2. Activated Cdk phosphorylates DNA polymerase and activates it;

Comments:

(d) Describe the role of DNA polymerase in DNA replication.

LO 2(b); AOA [2]

Answers:

1. adds free deoxyribonucleoside triphosphates / dNTP / DNA nucleotides / deoxyribonucleotides to the 3'-OH end of the RNA primer / growing DNA strand;
2. catalyses formation of phosphodiester/phosphoester bond(s);
3. corrects mistakes in DNA by removing and replacing the mismatched nucleotides / bases;
4. removes the RNA primer from the 5' end of the leading strand / Okazaki fragment and adds dNTP to fill the gap;

Comments:

(e) Fig. 3.2 shows a linear chromosome undergoing DNA replication.

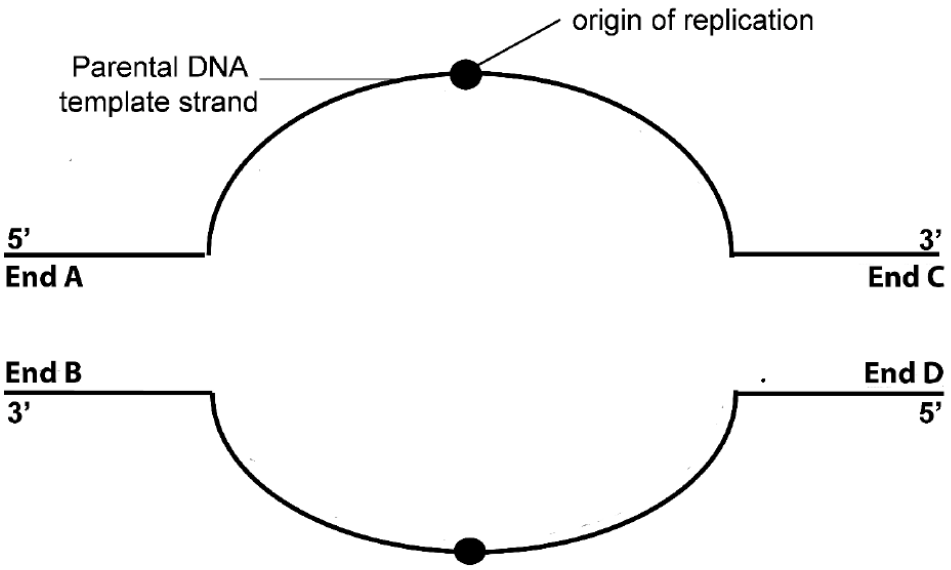


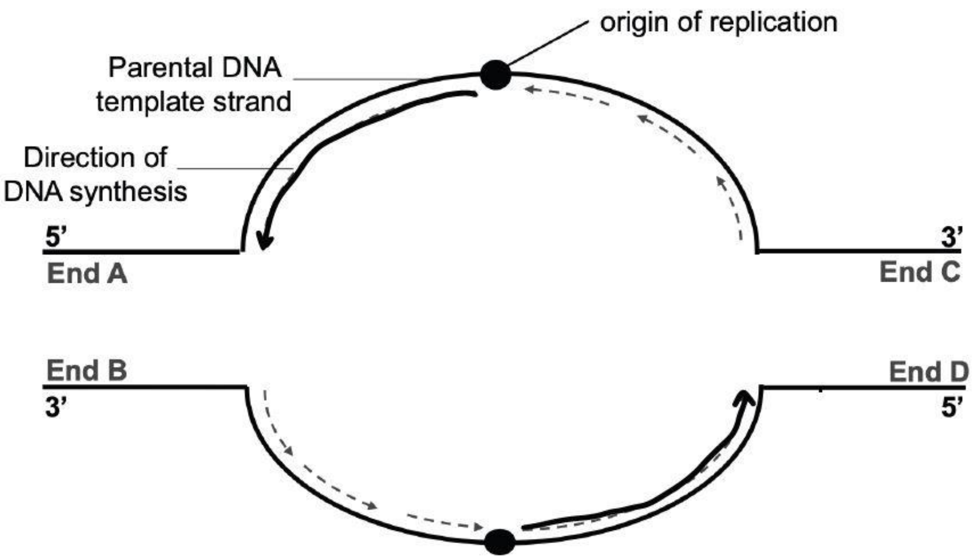
Fig. 3.2

(i) On Fig. 3.2, draw the direction of DNA replication for **one** leading (⟶) strand and **one** lagging (------▸) strand.

LO 2(b); AOA [2]

Answers:

1. just one correct leading strand direction drawn;
2. just one correct lagging strand direction drawn;



Comments:

(ii) Ends A, B, C and D represent the ends of the newly synthesised daughter DNA strands.

State which end(s) will experience the end-replication problem after the first round of DNA replication has been completed.

LO 2(b); AOA [1]

Answers:

1. End B and End C;

Comments:

[Total: 10]

4 Fig. 4.1 shows the pathways that can be activated by glucagon to increase the blood glucose level when it is bound to its receptor on the liver cell surface.

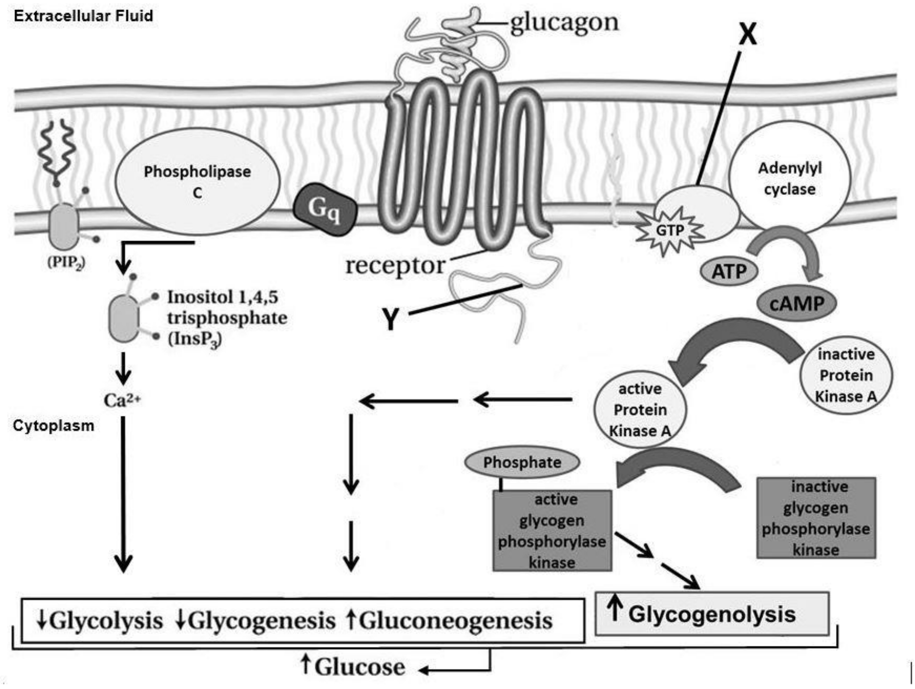


Fig. 4.1

(a) Name molecule **X** in Fig. 4.1.

LO 3(p); AOB [1]

Answers:

1. (activated) G_{α} subunit / α-subunit / GTP-binding protein / G protein;

Comments:

(b) With reference to Fig. 4.1,

(i) suggest the function of region **Y** of the glucagon receptor.

LO 3(p); AOB [1]

Answers:

1. Region **Y** is the G-protein binding site that allows the activated G-protein linked receptor / glucagon receptor to interact / bind with the G-protein / **X**;
2. Region **Y** is the binding site that allows the activated G-protein linked receptor / glucagon receptor to interact / bind with other protein(s) in the cytoplasm;

Comments:

(ii) describe how glucagon increases the blood glucose level when it is bound to its receptor on the effector cell surface.

LO 3(m), 3(p); AOB [5]

Answers:

Any 3 of A:
A1. Glucagon binds to the extracellular ligand-binding site of its receptor / G-protein linked receptor (GPLR), which then undergoes a conformational change that activates it;
A2. Activated GPLR binds to the inactive G-protein / **X /** G_q, which then undergoes a conformational change that causes its GDP bound to be displaced by GTP, forming an activated G-protein / **X /** G_q;

A3. Activated G-protein / **X** binds to membrane-bound adenylyl cyclase and activates it OR Activated G-protein / G_q binds to membrane-bound phospholipase C and activates it;

A4. Activated adenylyl cyclase catalyses the conversion of ATP into cAMP OR Activated phospholipase C catalyses the production of $InsP_3$ (from the hydrolysis of PIP_2);

A5. cAMP binds to Protein Kinase A and activates it OR $InsP_3$ causes intracellular Ca^{2+} release from the ER;

A6. Active Protein Kinase A phosphorylates (inactive) glycogen phosphorylase kinase and activates it;

Any 1 of B:
B1. Active glycogen phosphorylase kinase (indirectly) results in an increased rate of glycogenolysis;

B2. Active Protein Kinase A (indirectly) / Ca^{2+} causes decreased rate of glycolysis / decreased rate of glycogenesis / increased rate of gluconeogenesis;

Compulsory
C1. (compulsory point) The glucose formed is then released from the liver cell to increase the blood glucose level;

Note:
There are many types of G proteins, of which Gsα and Gq are related to the glucagon receptor. When Gsα is activated, adenylate cyclase is activated and intracellular cAMP production is increased, which in turn activates protein kinase A (PKA), and leads to phosphorylation of functional proteins in the cells. This pathway is called PKA pathway. When Gq is activated, the phospholipase

C-inositol triphosphate channel (PLC-INSP3) pathway can be activated, causing intracellular Ca2+ release. The above two pathways will directly or indirectly cause a decrease in glycolysis, a decrease in glycogen synthesis, an increase in gluconeogenesis, an increase in glycogenolysis, and eventually an increase in blood glucose.
C2.

Comments:

(c) A group of liver cells has been found to contain a mutation in a gene coding for phosphatase **W**. Phosphatase **W** acts on a particular substrate in the glucagon signalling pathway to prevent glycogenolysis.

With reference to Fig. 4.1, suggest the identity of the substrate of phosphatase **W** and explain how glycogenolysis is prevented.

LO 3(o), 3(p); AOB [3]

Answers:

1. glycogen phosphorylase kinase;
2. Phosphatase **W** removes the phosphate group from glycogen phosphorylase kinase, causing it to be in its inactive form;
3. Inactive glycogen phosphorylase is unable to phosphorylate the next kinase in line, thereby <u>preventing further signal transduction/ OWTTE</u> that leads to glycogenolysis; [no mark for just saying "glycogenolysis is prevented"]

Comments:

[Total: 10]

5 Induced pluripotent stem cells (iPSCs) have become a model system for studying tissue development and related human diseases, holding great promise for cell therapy. However recent studies find genetic abnormalities in iPSCs, which may harbour risk of tumorigenesis.

Fig. 5.1 shows the karyotype of an abnormal iPS cell line that is maintained in a research lab.

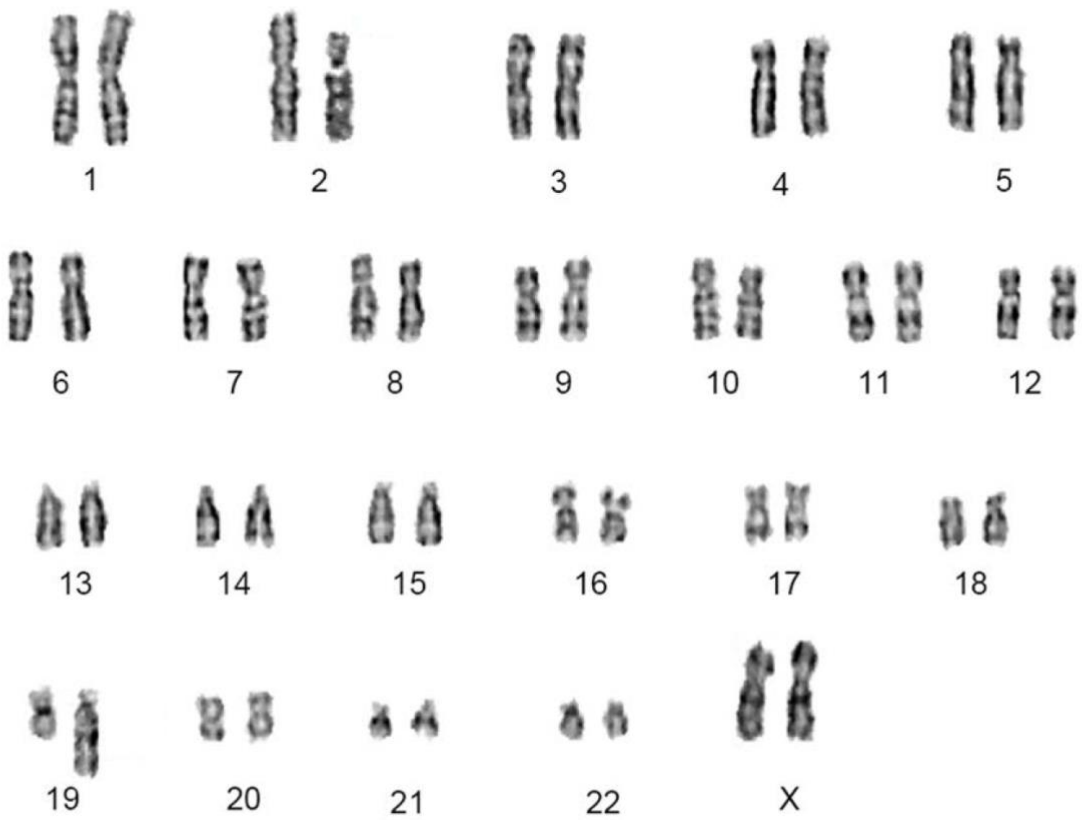


Fig 5.1

(a) Describe the genetic abnormality shown in Fig. 5.1

LO2l; AOB

Answers:

1. Ref to chromosomal aberration + translocation;

Any of:

2. Ref to part of chromosome 2 joining to chromosome 19;
3. Ref to one chromosome 2 shorter than other/ one chromosome 19 longer than other;
4. Ref to pairs of chromosomes 2 and 9 not appearing to be homologous;

Comments:

Fig 5.2 shows the lineage scheme of the somatic cell that was transformed into iPSC.

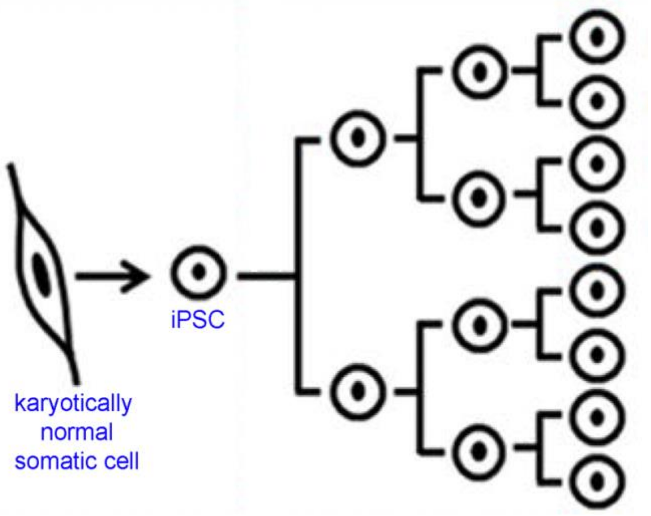


Fig. 5.2

(b) The genetic abnormality was found in approximately 53.3% of the progeny cells. All genetically abnormal iPSCs in the cell line carried the same karyotype.

(i) On Fig. 5.2, indicate and label where the genetic abnormality occurred. [1]

LO2n; POS2.6

Answers:

1. Ref to chromosomal aberration occurring during the first cell division;

Comments:

(ii) Outline the events that resulted in occurrence of genetic abnormality in 53.3% of the progeny.

LO2n; AOA [4]

Answers:

1. Ref to cell undergoing mitotic cell cycle before abnormality occurred;
2. Ref to translocation occurring before S phase;

3. Ref to S phase + replication of abnormal chromosomes;

4. Ref to M phase + cytokinesis + two genetically identical daughter cells;

5. Ref to abnormal cell undergoing repeated mitotic divisions to result in ~53.3%;

Comments:

(c) It is hypothesised that as somatic reprogramming to iPSCs undergoes a rapid increase in the rate of cell division, it leads to an increase in energy demand. The metabolic transformation causes a significant increase in the level of reactive oxygen species, resulting in oxidative DNA damage.

(i) Name another cause of DNA damage.

LO2p; AOA [1]

Answers:

1. Ionising radiation/ chemical carcinogen;

[Reject: radioactivity, exonucleases, end-replication problem, extreme temperatures, telomerases]

(ii) Suggest how DNA damage in the iPSC led to the karyotype shown in Fig. 5.1.

LO2o; AOB [2]

Answers:

1. Ref to double stranded breaks in chromosome 2 and 19;

2. Ref to activation of cell cycle checkpoint before S phase;

3. Ref to ligation of DNA of chromosome 2 and chromosome 19;

4. Ref to colocation of chromosome 2 and chromosome 19 in the nucleus during interphase;

Comments:

[Total: 10]

6 Tomato varieties vary in many characters, such as stature (tall versus dwarf) and skin structure (smooth versus peach skin). Plants from F_1 generation which was tall and had smooth fruits were self-pollinated.

In the F_2 generation: 101 plants were tall and had smooth fruits : 56 plants were tall and had peach fruits: 52 plants were dwarf and and smooth fruits : none were dwarf and had peach fruits.

(a) Draw a genetic diagram to explain the distribution of plants in the F_2 generation.

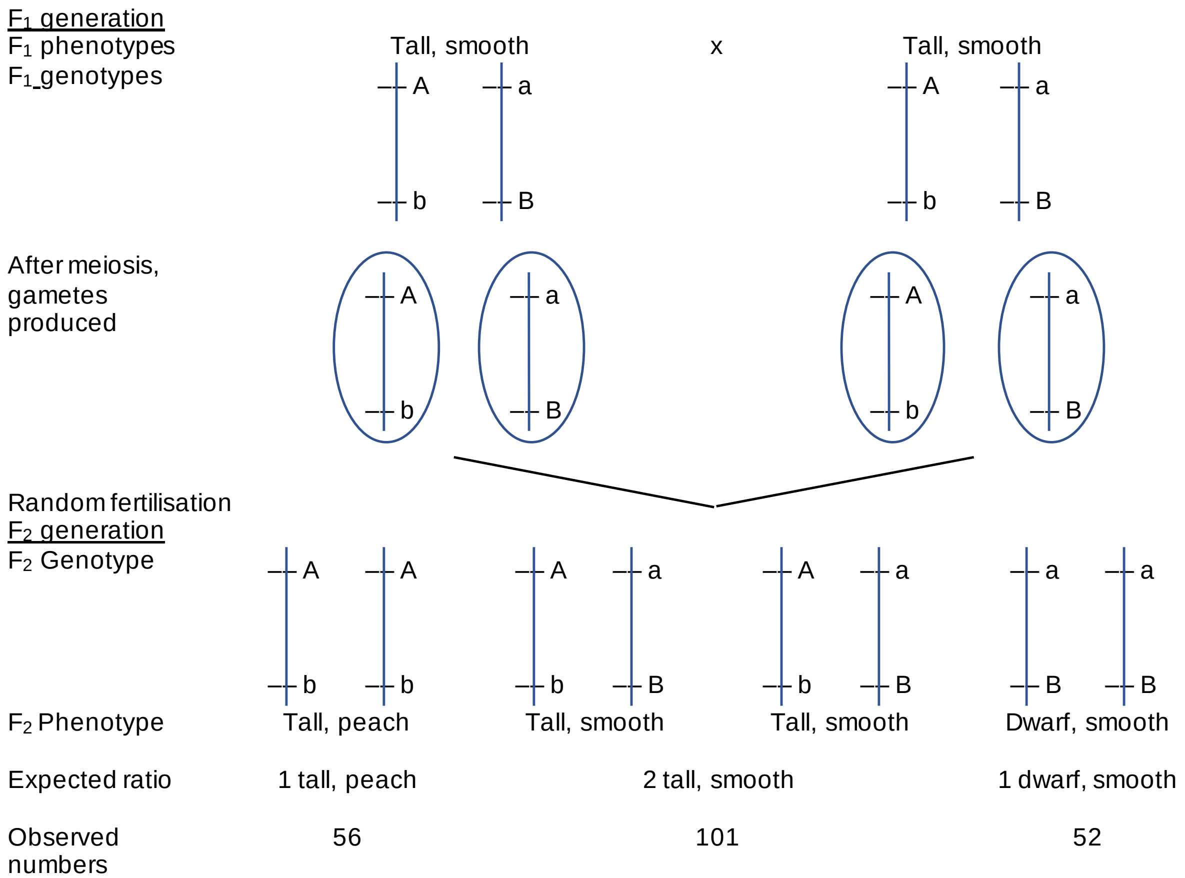
Use the symbols **A**, **a** and **B**, **b** to represent the alleles.

LO2z; AOB [4]

Answers:

1. represents dominant and recessive alleles for stature and skin structure (accept A/B for stature and B/A for skin structure);
2. represents F1 generation with linkage;
3. represents gametes in circles (accept if student did not show linkage) + gametes should be logical wrt F1 phenotype;
4. represents the F2 generation genotype and phenotype appropriately (reject if no linkage) ;
5. relate expected ratio to observed numbers ;

Let:
A represent dominant allele for tall, **a** represent recessive allele for dwarf
B represent dominant allele for smooth, **b** represent recessive allele for peach



Comments:

(b) Carry out chi-squared test to support your answer in **(a)**.

(i) State the null hypothesis.

LO2dd; AOA [1]

Answers:

1. There is no significant difference between observed numbers and expected numbers based on a dihybrid inheritance. Any difference is due to chance;

Comments:

(ii) Complete the table.

LO2dd; AOA

Answers:

1. State all expected numbers in whole numbers correctly;
2. Complete the (O-E)^2/E appropriately + present calculations to at least 2 dp;
3. Complete the final summation correctly + present answer to 2dp or 3sf;

	Observed numbers (O)	Expected numbers (E)	$\frac{(O-E)^2}{E}$
tall smooth	101	118	2.4491
tall peach	56	39	7.4102
dwarf smooth	52	39	4.3333
dwarf peach	0	13	13
		$\chi^2 = \sum \frac{(O-E)^2}{E}$	27.19(26)

[3]

Comments:

Expected numbers should be presented in whole numbers.

(iii) Critical values of this chi-squared distribution are shown in Table 6.1.

Table 6.1

degrees of freedom	probability								
	0.99	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.01
2	0.02	0.10	0.21	0.58	1.39	2.77	4.61	5.99	9.21
3	0.12	0.35	0.58	1.21	2.37	4.11	6.25	7.81	11.34
4	0.30	0.71	1.06	1.92	3.36	5.39	7.78	9.49	13.28

State the conclusions that may be drawn from **(b)(ii)**.

LO2dd; AOA [2]

Answers:

1. calculated χ^2 value, 27.19 is greater critical value, 7.81 at p=0.05;
2. observed numbers deviate significantly from the expected ratio of 9:3:3:1;
3. reject null hypothesis;
4. there was no independent assortment of the two genes;

Comments:

[Total: 10]

7 Fig. 7.1 shows the reduction of NAD during cellular respiration.

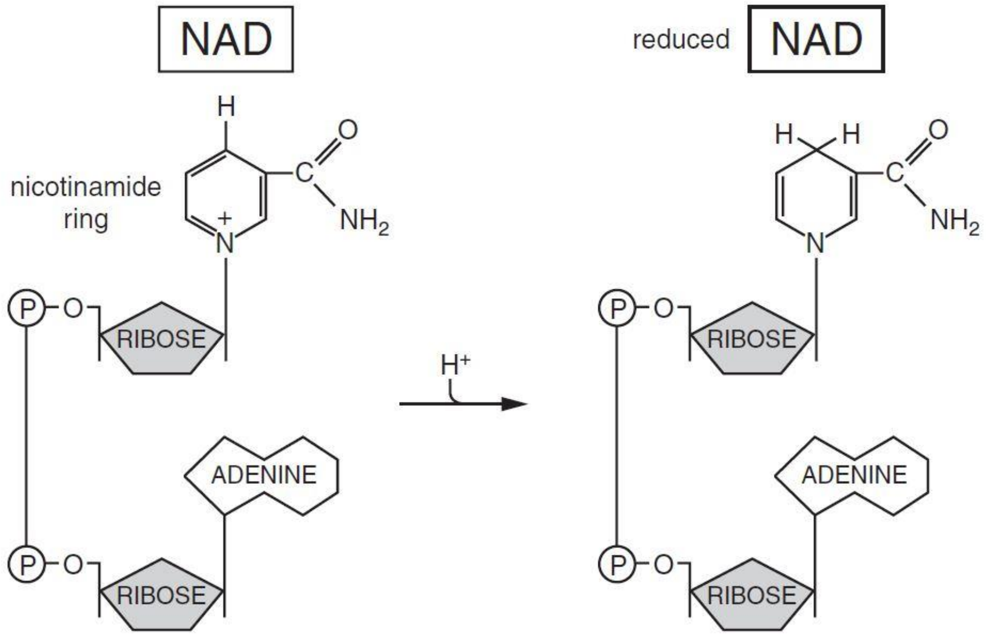


Fig. 7.1

(a) State **two** locations in the eukaryotic cell where NAD is reduced.

LO 3(f), 3(g); AOA [2]

Answers:

1. cytoplasm;
2. mitochondrial matrix;

Comments:

(b) Outline the role of NAD in cellular respiration.

LO 3(f), 3(g), 3(h); AOA [3]

Answers:

1. coenzyme / electron carrier
2. NADH / reduced NAD carries electrons / protons / hydrogen ions / hydrogen / H / 2H / H^+; (reject: H_2)
3. to **<u>electron transport chain</u>** from glycolysis / link reaction / Krebs cycle;

4. role of NAD^+ / oxidised NAD in conversion / oxidation of triose phosphate to pyruvate in glycolysis;

5. role of NADH / reduced NAD in (alcoholic / lactic acid) fermentation / anaerobic respiration;

Comments:

(c) Explain why NAD cannot be regenerated from reduced NAD in mitochondria in the absence of oxygen.

LO 3(h); AOA [3]

Answers:

1. electron transport chain stops functioning in the absence of oxygen;

2. role of oxygen as the final electron acceptor at the end of electron transport chain;

3. reduced NAD cannot be oxidised / electrons cannot be passed from NADH to the ETC

(d) Yeast can respire aerobically and anaerobically. When there is insufficient oxygen, yeast cells switch from aerobic to anaerobic respiration. This results in a significant increase in the rate of glucose uptake and glycolysis in the yeast cells.

Explain why the rate of glycolysis increases significantly when yeast cells switch from aerobic to anaerobic respiration.

LO 3(f), 3(h), 3(i); AOB [2]

Answers:

1. aerobic respiration produces more ATP / anaerobic respiration produces fewer ATP;

2. **more glucose** needs to be broken down in glycolysis to **produce the same amount of ATP**; R: produce **more** ATP

3. no electron transport chain / oxidative phosphorylation during anaerobic respiration;

[Total: 10]

8 Fig. 8.1 shows the development of myeloid and lymphoid cell types from haematopoietic blood stem cell.

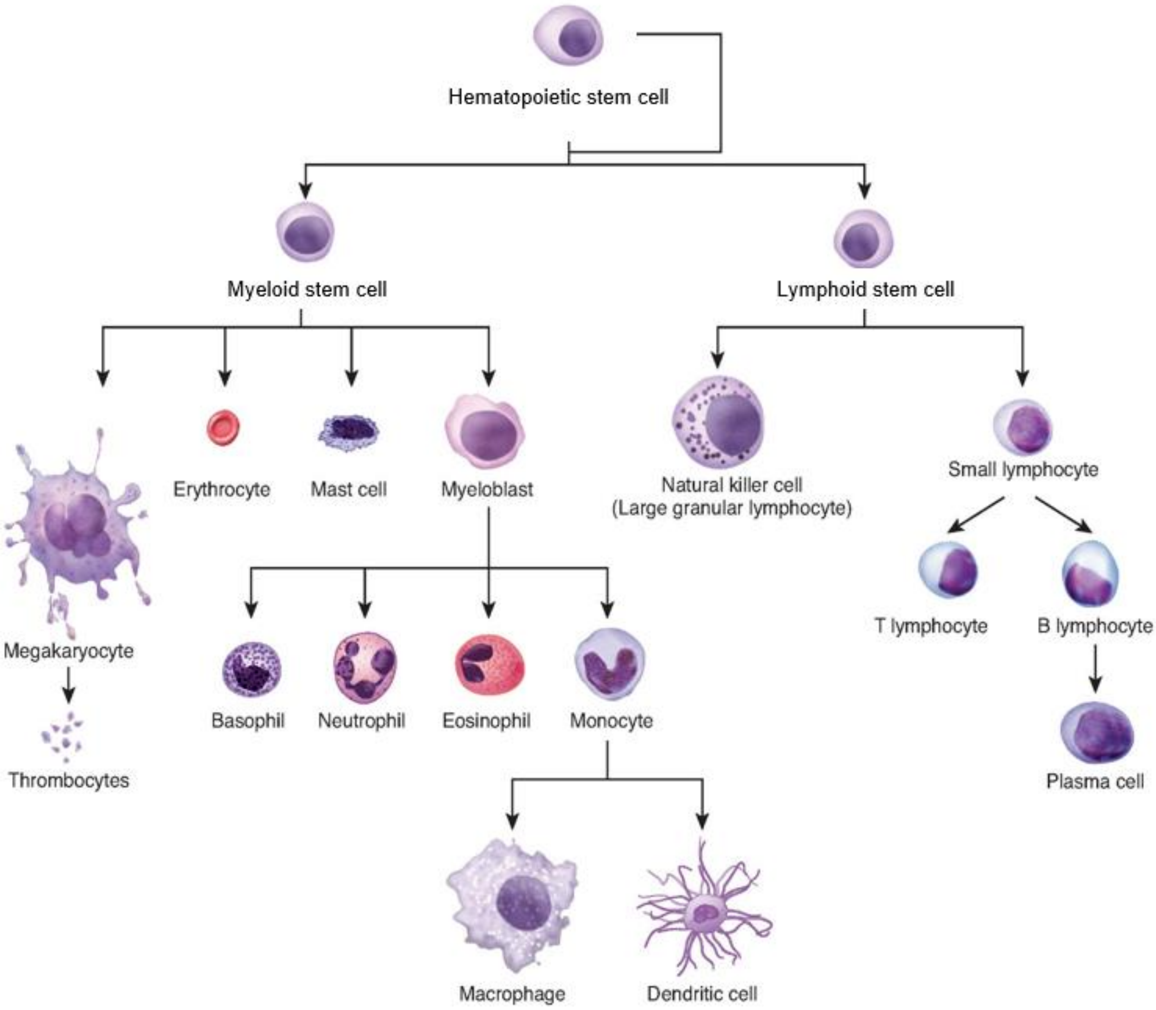


Fig. 8.1

(a) With reference to Fig. 8.1, describe the features and functions of lymphoid stem cell.

LO1t; AOA [3]

Answers:

1. Lymphoid stem cells are **multipotent**
2. Can divide and **self-renew indefinitely** to produce/replace more lymphoid stem cells R: infinite / divides indefinitely
3. Cells are **unspecialised** and
4. have the ability to **differentiate** into specialised cell**s e.g. natural killer cell, small lymphocyte** (at least 1 example)

Natural killer cell deficiency (NKD) is a primary immunodeficiency where the main defect lies in NK cells.

One possible cause of NKD is the *GATA2* mutation.

GATA2 is a transcription factor that regulates the proliferation, differentiation and survival of blood stem cells. GATA2 has a zinc finger domain that allows for DNA-protein interaction. Fig. 8.2 shows the zinc finger interacting with a segment of DNA.

Fig. 8.2

(b) Explain how the zinc finger domain can bind to the DNA double helix.

LO2j: AOB [3]

Answers:

1. Zinc finger has a **tertiary structure** with **3D conformation**
2. **Complementary in shape and charge** to the **DNA** double helix
3. Presence of **positively charged R groups of amino acid** that can interact with the **negatively charged DNA**
4. Zn2+ forms **ionic interactions/electrostatic forces of attraction** with negatively charged DNA

Comments:

(c) Natural Killer (NK) cells are innate immune cells that show strong cytolytic functions against physiologically stressed cells such as tumour cells and virus-infected cells.

Suggest how GATA2 mutation may lead to increased incidence of cancer.

LO Extension A: AOB [4]

Answers:

1. Failure of GATA2 TF to bind to DNA of blood stem cells **prevents differentiation into NK Cells**

2. Loss of immunity

3. Ref: Decrease in population of NK Cells to bind to tumour cells

4. Ref: reduction in cytolytic ability of immune system to remove tumour cells, hence increase chance of cancer

Comments:

[Total :10]

9 To study the evolution of sleep, scientists at the European Molecular Biology Laboratory in Germany study the activity of genes involved in making the sleep hormone, melatonin, and the melatonin activity in the marine worm, *Platynereis dumerilii*.

Fig. 9.1 shows a two-day old larva of *P. dumerilii*. The larva has locomotor-ciliated cells that regulates vertical swimming.

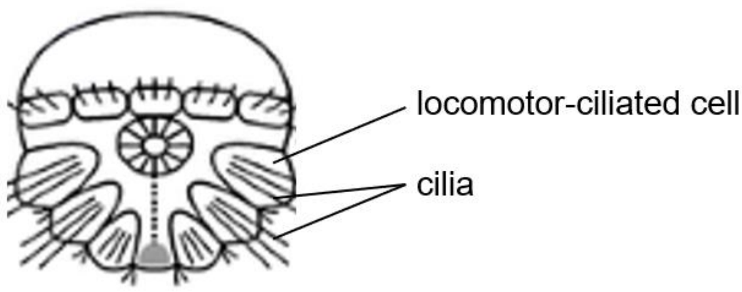


Fig. 9.1

During light hours, melatonin production is inhibited. During dark hours, melatonin is produced. Melatonin induces rhythmic burst firing by motor neurons to release acethylcholine (ACh) that targets locomotor-ciliated cells. Ciliary arrests were found to be triggered by ACh transmission.

Fig. 9.2 shows the interaction between light-dark hours, melatonin production and regulation of vertical swimming.

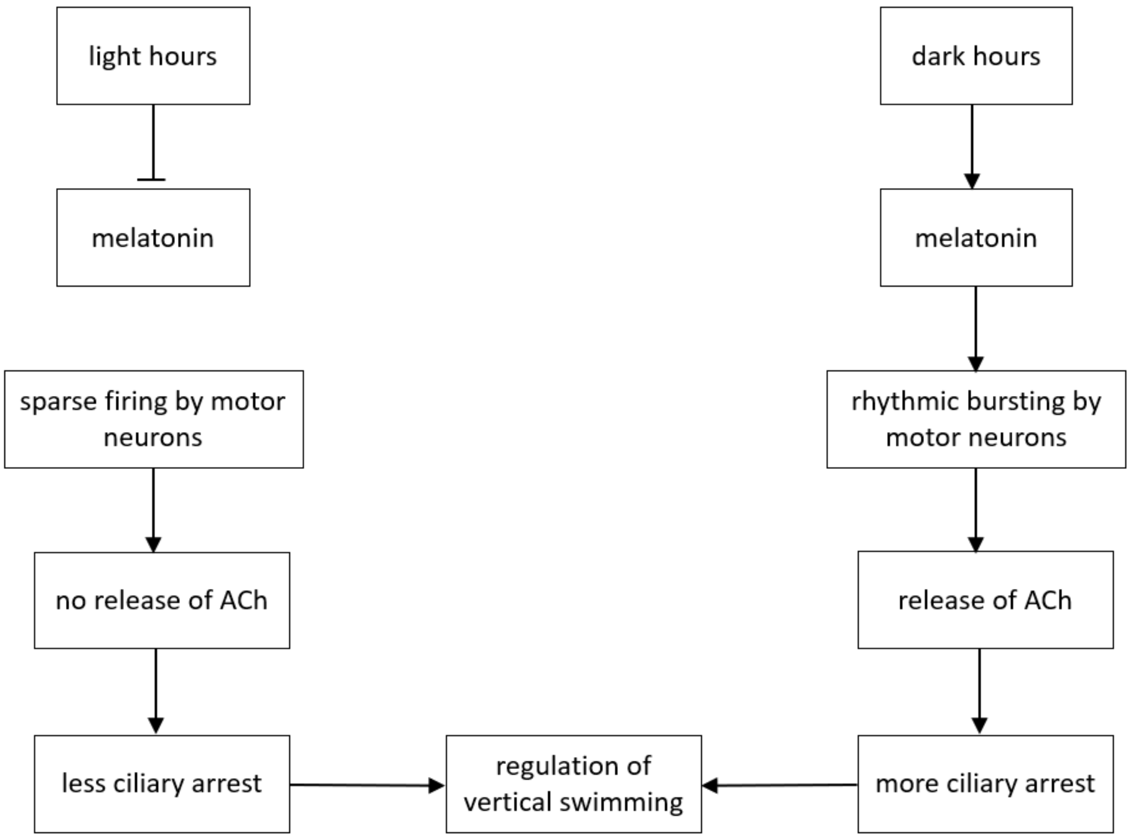


Fig. 9.2

(a) With reference to Fig. 9.2, suggest and explain whether the larva will be near the surface or bottom of the water body when there is light.

LO POS; AOB [3]

Answers:

1. surface
2. Inhibit the production of melatonin → Sparse firing by motor neurons
3. leading to no release of ACh → less ciliary arrests

Comments:

(b) Suggest and explain **one** evolutionary advantage of the reaction to the sleep hormone for the marine worm, *P. dumerilii.*

LO4 POS: AOB [2]

Answers:

1. Energy saving
2. Worm larvae only need to swim up vertically during the day / at night the worm can rest
3. Avoiding predators
4. By sinking to the bottom of the ocean/sea

Comments:

(c) The biochemical pathway of melatonin synthesis is shown in Fig. 9.3.

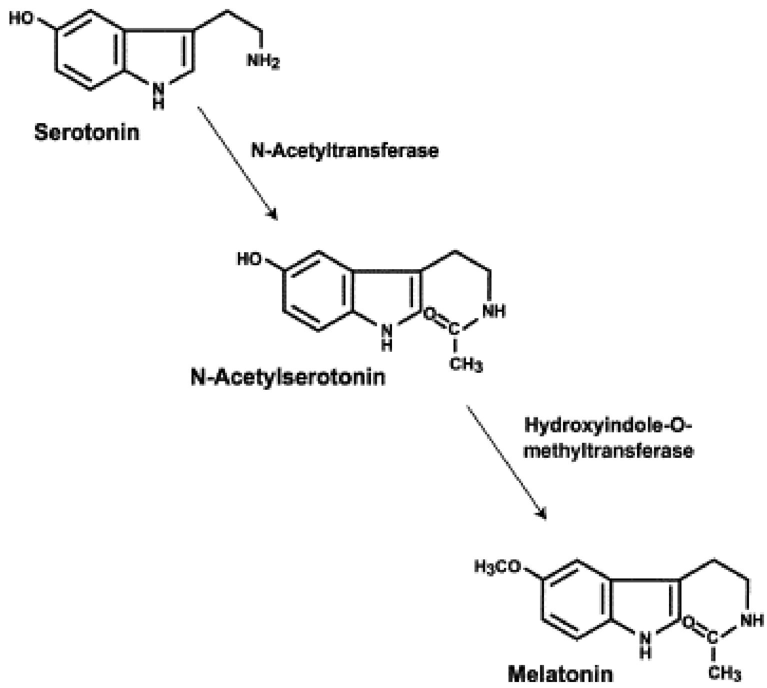


Fig. 9.3

The process to synthesize melatonin from serotonin requires two enzymes, N-acetyltransferase and HydroxyIndole-O-methyltransferase. Recessive alleles that are not able to code for either of the enzyme have been maintained in the population of *P. dumerilii*.

Explain how the recessive allele may be maintained in the population of *P. dumerilii*.

LO 4e: AOB [2]

Answers:

1. Diploidy/heterozygote protection

2. Effect of the dominant allele masks the effect of the recessive allele

Comments:

(d) *P. dumerilii* are considered living fossils as fossils of this organism have been found as far back as the Cambrian era (about 500 million years ago). *P. dumerilii* are thought to be the last common ancestor of almost all living animals, including worms, flies and humans. A population of *P. dumerilii* developed photoreceptor cells and ability to produce and react to melatonin.

Explain why population is the smallest unit that can evolve.

LO2d: AOA [3]

Answers:

1. **Individuals** do not evolve, but **can accumulate mutations / no change in allele frequency**
2. Evolution is the **change of allele frequency** within a population
3. **Variation exists in a population** and **natural selection acts on the individuals/alleles** in a population R: selection acts on traits
4. Selecting **advantageous alleles** to be **passed into successive generation** R: advantageous phenotypes/traits
5. Over many generations, a **higher proportion** of the population will bear the selected alleles

Comments:

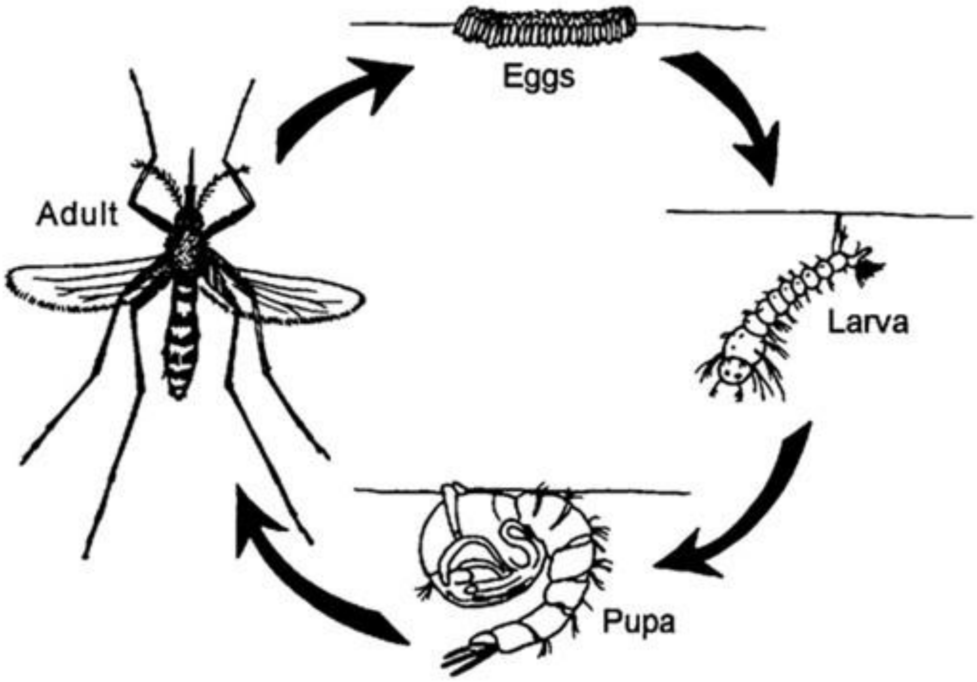
[Total :10]

10 Dengue viruses are spread to people through the bites of infected *Aedes aegypti*. These mosquitoes typically lay eggs near standing water in containers that hold water.

(a) Using the space below, draw a labelled diagram of the life cycle of the *A. aegypti* mosquito and indicate the duration of each stage.

LOB(f) [2]

Answers:



Source: NEA (https://www.nea.gov.sg/docs/default-source/resource/section-a---mosquitoes-amp-mosquito-borne-diseases-pdf-1-41-mb-.pdf)

1. Eggs to larva (2-3 days)
2. Larva to pupa (4-5 days)
3. Pupa to adult (1-2 days)
4. Adult lay eggs after blood meal (3-4 days)

Source: notes

1. Eggs to larva (2-7 days)
2. Larva to pupa (more than 4 days)
3. Pupa to adult (2 days)

Fig. 10.1 shows the prevalence of dengue virus outbreak and flushing in Singapore for a typical year. Flushing is the surface runoff during heavy rainfall.

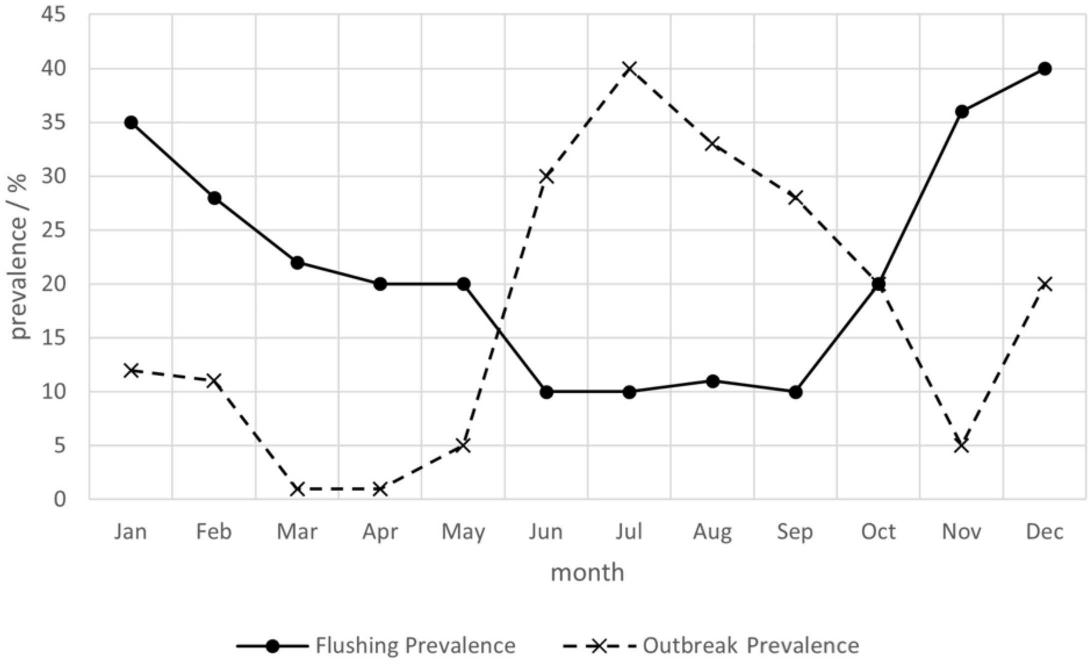


Fig. 10.1

(b) With reference to Fig. 10.1,

(i) describe how flushing prevalence changes over the months.

LO: ; POS [2]

Answers:

1. flushing prevalence decreases from 35% in January to about 10% in June
2. Stays roughly constant at 10-13% from June to Sep
3. Increases to 40% in Dec

Comments:

(ii) suggest and explain how flushing affects outbreak prevalence.

LO: B(f); POS [2]

Answers:

1. Quote data: e.g. when flushing prevalence is low at 10% from June to Sept, Outbreak prevalence is high, peaking at 40% in Jul;

2. Less stagnant water results in less water for eggs to develop

R: kill mosquito / eggs; need to explain how

Comments:

(c) Draw a line in Fig. 10.1 to show how global warming can affect outbreak prevalence. [1]

LO: B(e); POS

Answers:

1. Higher peak for outbreak prevalence
2. Earlier peak (at least 40%)

(d) Describe and explain what is shown by the line you have drawn in Fig. 10.1.

LO: POS B(e) [2]

Answers:

1. General: Outbreak prevalence increases for **every month** compared to current
2. Higher peak for outbreak prevalence
3. Ref: Due to increase in mean temperature in Jul, increase rate of mosquito reproducing / dengue virus replicates faster → faster spread of dengue virus
4. Earlier peak for outbreak prevalence
5. Ref: Due to increase in mean temperature earlier in the year, mosquito starts reproducing faster earlier in the year.

(e) Explain why an individual can contract dengue more than once.

LO: A (b) and B(g) [2]

Answers:

1. Immunological memory present if the serotype of dengue is the same in primary and secondary infection is the same.
2. No immunity if serotype is different / mutation of serotype
3. Memory cells are activated and differentiate into effector cells, e.g. cytotoxic T-cells and plasma cells but are not effective.

[Total: 10]